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Course: Calculus Beta

Assignment: MyLab Math - Uge 12

Use these equations to find the values of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at the given point.

$$4z^3 - 2xy + 5yz + 3y^3 - 34 = 0, (1, 2, 1)$$

If the equation $F(x, y, z) = 0$ determines z as a differentiable function of x and y , then, at the points where $F_z \neq 0$, the following equations are true.

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \text{ and } \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

To find the value of $\frac{\partial z}{\partial x}$ at the point $(1, 2, 1)$, first determine $\frac{\partial z}{\partial x}$ by calculating F_x , the partial derivative of the given function with respect to x , by treating y and z as constants. Then calculate F_z , the partial derivative of the given function with respect to z , by treating x and y as constants.

Calculate each partial derivative, F_x and F_z , given that $F(x, y, z) = 4z^3 - 2xy + 5yz + 3y^3 - 34$.

$$F_x = -2y$$

$$F_z = 12z^2 + 5y$$

Substitute $-2y$ for F_x and $12z^2 + 5y$ for F_z to find $\frac{\partial z}{\partial x}$.

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}$$

$$\frac{\partial z}{\partial x} = -\frac{-2y}{12z^2 + 5y}$$

Substitute the partial derivatives accordingly.

$$\frac{\partial z}{\partial x} = \frac{2y}{12z^2 + 5y}$$

Simplify.

Now substitute the coordinates of the given point, $(1, 2, 1)$, for x , y , and z , then simplify to find the value of $\frac{\partial z}{\partial x}$.

$$\left. \frac{\partial z}{\partial x} \right|_{(1, 2, 1)} = \frac{2(2)}{12(1)^2 + 5(2)}$$

Substitute 1 for x , 2 for y , and 1 for z .

$$\left. \frac{\partial z}{\partial x} \right|_{(1, 2, 1)} = \frac{2}{11}$$

Simplify.

Therefore, the value of $\frac{\partial z}{\partial x}$ at the point $(1, 2, 1)$ is $\frac{2}{11}$.

To find the value of $\frac{\partial z}{\partial y}$ at the point $(1, 2, 1)$, begin by calculating F_y , the partial derivative of the given function with respect to y , by treating x and z as constants. Then calculate F_z , the partial derivative of the given function with respect to z , by treating x and y as constants.

Calculate each partial derivative, F_y and F_z , given that $F(x, y, z) = 4z^3 - 2xy + 5yz + 3y^3 - 34$.

$$F_y = -2x + 5z + 9y^2$$

F_z was previously determined and is shown below.

$$F_z = 12z^2 + 5y$$

Substitute $-2x + 5z + 9y^2$ for F_y and $12z^2 + 5y$ for F_z to find $\frac{\partial z}{\partial y}$.

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

$$\frac{\partial z}{\partial y} = -\frac{-2x + 5z + 9y^2}{12z^2 + 5y}$$

Now substitute the coordinates of the given point, $(1,2,1)$, for x , y , and z , then simplify to find the value of $\frac{\partial z}{\partial y}$.

$$\left. \frac{\partial z}{\partial y} \right|_{(1,2,1)} = -\frac{-2(1) + 5(1) + 9(2)^2}{12(1)^2 + 5(2)}$$

Substitute 1 for x , 2 for y , and 1 for z .

$$\left. \frac{\partial z}{\partial y} \right|_{(1,2,1)} = -\frac{39}{22}$$

Simplify.

Therefore, the value of $\frac{\partial z}{\partial y}$ at the point $(1,2,1)$ is $-\frac{39}{22}$.